HOTPOTQA

1.Overview

This document presents a concise, professional revision of the selected Word content describing HOTPOTQA, a large-scale dataset designed to evaluate multi-hop question answering (QA) and explainable reasoning over multiple documents.

Abstract

HOTPOTQA is a comprehensive dataset created to assess multi-hop question answering capabilities. Unlike conventional QA datasets that typically require information extraction from a single paragraph or document, HOTPOTQA mandates the aggregation of evidence from multiple Wikipedia articles to produce correct answers. The dataset comprises approximately 113,000 question-answer pairs, includes annotated supporting facts that justify each answer, and introduces comparison-style questions. Two benchmark settings are provided: Distractor and Full Wiki. Experimental evaluations demonstrate that current QA systems perform substantially worse than humans, establishing HOTPOTQA as a challenging benchmark for multi-document reasoning and explainability.

Purpose

The primary objective of HOTPOTQA is to enable rigorous evaluation of systems that must reason across documents rather than rely on single-paragraph extraction. Specifically, the dataset aims to address three limitations observed in prior QA resources: (1) the predominance of single-document reasoning tasks, (2) the reliance of many multi-hop datasets on structured knowledge bases, which reduces naturalness and variety, and (3) the absence of explicit supporting evidence that explains why an answer is correct. HOTPOTQA mitigates these issues by providing multi-document questions, human-annotated supporting facts, and targeted comparison questions to probe diverse reasoning skills.

Problems withExisting Datasets

Prior QA datasets exhibit several shortcomings that motivated HOTPOTQA:

- Single-document focus: Datasets such as SQuAD typically require only a single paragraph to locate an answer.
- Knowledge-base dependency: Resources like QAngaroo and ComplexWebQuestions depend heavily on structured knowledge bases, limiting question diversity and natural language variability.
- Lack of explainability: Most datasets evaluate only final-answer correctness without providing sentence-level evidence that supports the answer.
- Shallow reasoning evaluation: Numerous datasets emphasize retrieval and pattern matching rather than genuine multi-hop reasoning across documents.

5.DatasetOverview

Name: HOTPOTQA

Source: English Wikipedia

Yang, Zhilin, et al. "HotpotQA: A dataset for diverse, explainable multi-hop question answering." Proceedings of the 2018 conference on empirical methods in natural language processing. 2018.

Size: 113k

Main characteristics: Wikipedia-based, multi-hop QA, multi-document reasoning, annotated supporting facts, inclusion of comparison questions, and an explicit emphasis on explainable reasoning. The dataset is designed to require systems to produce both correct answers and the evidence used to derive them.

Dataset Structure

Each example in HOTPOTQA contains the following components:

- Question: The natural-language query posed to the system.
- Context: Two or more related Wikipedia articles that together contain the information required to answer the question.
- Answer: The correct response to the question.
- Supporting facts: Sentence-level annotations that justify the answer and expose the reasoning chain.

Creation Process

The dataset was created using a controlled pipeline to ensure that examples require multi-hop reasoning:

1. Extract an English Wikipedia dump.
2. Construct a hyperlink graph that models inter-article links.
3. Select related article pairs or small subgraphs for question generation.
4. Present these articles to crowd workers.
5. Instruct workers to author questions that require synthesizing information from multiple articles.
6. Have workers annotate sentence-level supporting facts that justify the answer.
7. Apply quality-control checks before adding examples to the final dataset.

Hyperlink Graph and Bridge Entities

A hyperlink graph represents Wikipedia as a directed graph: articles are nodes and hyperlinks are edges. This graph assists in locating related articles suitable for multi-hop question generation. Bridge entities are intermediate entities that connect disparate facts across articles and thereby enforce multi-step reasoning. For example, to answer a question about the birth date of a musician affiliated with a specific band, the model must identify the band, locate the musician (bridge entity), and then extract the musician’s birth date. Bridge entities thereby convert simple retrieval into a multi-hop reasoning task.

Statistics and Splits

Dataset splits and counts are provided to facilitate training and evaluation:

- Train (Easy): 18,089
- Train (Medium): 56,814
- Train (Hard): 15,661
- Development: 7,405
- Test (Distractor): 7,405
- Test (Full Wiki): 7,405
- Total: 112,779

Selected Task Details

Item	Details
Question Categories	People; Places; Dates; Events; Numbers; Yes/No; Comparison
Answer Types	Person≈30%; Organization/Group≈13%; Location≈10%; Date≈9%; Number≈8%; Artwork≈8%; Yes/No≈6%; Adjective≈4%; Event≈1%; Other Proper Nouns≈6%; Common Nouns≈5%
Purpose	Evaluate diverse reasoning and extraction scenarios

Benchmark Settings and Results

Two evaluation settings are defined:

- Distractor: Each example supplies two gold paragraphs and eight distractor paragraphs; the model must find the answer and supporting facts among these passages.
- Full Wiki: The model must retrieve relevant documents from the entire Wikipedia, locate answers, and identify supporting facts. This setting adds a document-retrieval challenge and is substantially more difficult.

Task	Metric	Value
Answer	EM	≈45.46
Answer	F1	≈58.99
Supporting Fact	EM	≈22.24
Supporting Fact	F1	≈66.62
Joint	EM	≈12.04
Joint	F1	≈41.37

Limitations

•Document retrieval is difficult in the Full Wiki setting.
•Automated systems lag behind humans on multi-hop reasoning.
•Predicting supporting facts precisely remains challenging even when answers are correct.
•Reliance on Wikipedia limits domain coverage and breadth

15.Conclusion

HOTPOTQA represents a rigorous benchmark for multi-hop, explainable question answering. By combining large scale, human-annotated supporting facts, and settings that vary retrieval difficulty, it provides a valuable resource for advancing research on multi-document reasoning, interpretable predictions, and robust document retrieval.